Mohit Singh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	6.97
Hansraj modern school	Class XII (CBSE)	2024	90.00%
Kendriya vidhyalaya No.3	Class X (CBSE)	2022	91.00%

SKILLS

Programming Languages: HTML, CSS, C, C++, Python, Verilog, MATLAB.

EDA & Design Tools : Vivado, LTSpice, KiCad, Cadence Virtuoso, EasyEda pro.

Python Libraries: NumPy, Matplotlib.

OS: Windows, Ubuntu , Arch.

Other Tools: Git, LaTeX, Arduino IDE, PlatformIO.

RELEVANT COURSES

Computer Science Courses: Data Structures and Applications, Introduction to Programming .

Math Courses: Mathematics for EE, Matrix Theory, Probability, Differential Equations and Transforms, Random Processes .

Core Courses: Digital Systems, Machine Learning, Analog Electronics, Digital Signal Processing, Control Systems, Semiconductor Device Fundamentals, Circuits and Network Analysis.

PROJECTS

Microphone Amplifier Design

- Designed a **two-stage audio amplifier** consisting of an **LM358 preamplifier** and **Class-AB BJT power stage** for driving an **8 speaker**.
- Optimized the amplifier through **iterative biasing, gain, and noise improvements**, including a **0-18 V/V variable-gain preamplifier** and volume control.
- Characterized the design by measuring **gain, bandwidth, THD, PSRR, quiescent current, output swing, and noise**, achieving **15.66 dB gain**, **15.2 kHz bandwidth**, and **4.3% THD**.

6-bit SAR ADC with R-2R DAC (Arduino-based)

- Designed and implemented a **6-bit SAR ADC** featuring a discrete **R-2R ladder DAC**, comparator, and Arduino-based SAR control logic.
- Characterized ADC performance by analyzing **quantization error, sampling theorem, and aliasing** over a **150 Hz-2 kHz** sampling range.
- Validated ADC functionality through **LED-based digital output** and **real-time serial acquisition** of conversion data for waveform visualization and analysis.

Embedded System

- **Scientific Calculator :** Designed and developed a **scientific calculator** using an **Arduino Uno**, a **26-button matrix keypad**, and a **LCD**, implementing arithmetic and scientific functions through embedded programming, integrating hardware peripherals, and validating functionality via breadboard testing.
- **Digital Clock :** Built a **digital clock** using an **Arduino Uno**, six **7-segment displays**, and a **7447 BCD-to-7-segment decoder**, implementing manual time-setting controls and validating real-time clock operation

through breadboard-based hardware testing.

QR Algorithm with Householder Transformations & Wilkinson Shift

- Implemented a QR eigenvalue solver in C using **Householder reflections** and a **Wilkinson-shifted QR iteration** for convergence, handling complex-conjugate eigenvalues via 2×2 Hessenberg block resolution with $O(n^3)$ time complexity.

Mini Projects

- **8-bit Asynchronous Counter:** Designed and verified an **8-bit ripple counter** using flip-flops, analyzing timing and propagation delay.
- **Finite State Machine (FSM):** Designed and implemented **Moore and Mealy FSMs** for sequential logic applications.
- **3D Modeling & Printing:** Designed CAD models in **FreeCAD** and fabricated functional prototypes using a **Creality 3D printer**.
- Implemented **DFT/IDFT** from scratch, verified core DFT properties against NumPy's **FFT** (**~11,400x speedup** benchmark), and applied **frequency-domain filtering** and **reverberation/dereverberation** to real audio signals.
- **RTL Design (Verilog)** -Implemented a synchronous **FIFO buffer** in **Verilog** with **circular-buffer addressing**, **occupancy tracking**, and **status flags** for full/empty conditions.
- **NeuraCore -- CNN Inference Accelerator Development board using Zynq-7000 SoC** (Ongoing)-Designing the schematic and PCB layout for the NeuraCore development board in KiCad.

POSITOINS OF RESPONSIBILITY

Teaching Assistant | EE1030: Matrix Analysis, IIT Hyderabad - Evaluated C/Python programming assignments and mentored 1st-year undergraduate students in the hardware laboratory.

EXTRACURRICULAR

National Sports Organization (NSO), IIT Hyderabad - Active participant in volleyball.